

C Programs for Common Problems

Matrix Multiplication

```
#include <stdio.h>

void multiplyMatrices(int first[][10], int second[][10], int result[][10], int r1, int
c1, int r2, int c2) {
    if (c1 != r2) {
        printf("Matrices cannot be multiplied!
");
        return;
    }
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++) {
            result[i][j] = 0;
            for (int k = 0; k < c1; k++) {
                result[i][j] += first[i][k] * second[k][j];
            }
        }
    }
}

int main() {
    int r1, c1, r2, c2;

    int first[10][10], second[10][10], result[10][10];

    printf("Enter rows and columns of first matrix: ");

    scanf("%d %d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
```

```
scanf("%d %d", &r2, &c2);

printf("Enter elements of first matrix:

");

for (int i = 0; i < r1; i++) {

    for (int j = 0; j < c1; j++) {

        scanf("%d", &first[i][j]);

    }

}

printf("Enter elements of second matrix:

");

for (int i = 0; i < r2; i++) {

    for (int j = 0; j < c2; j++) {

        scanf("%d", &second[i][j]);

    }

}

multiplyMatrices(first, second, result, r1, c1, r2, c2);

printf("Resultant Matrix:

");

for (int i = 0; i < r1; i++) {

    for (int j = 0; j < c2; j++) {

        printf("%d ", result[i][j]);

    }

    printf("\n");

}

return 0;

}
```

Bubble Sort

```
#include <stdio.h>

void bubbleSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main() {
    int n, arr[100];

    printf("Enter the size of the array: ");

    scanf("%d", &n);

    printf("Enter the elements of the array: ");

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    bubbleSort(arr, n);

    printf("Sorted Array: ");

    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
}
```

```
    return 0;
}
```

Binary to Decimal Conversion

```
#include <stdio.h>

int binaryToDecimal(int n) {
    int decimal = 0, base = 1, rem;
    while (n > 0) {
        rem = n % 10;
        decimal += rem * base;
        base *= 2;
        n /= 10;
    }
    return decimal;
}

int main() {
    int binary;

    printf("Enter a binary number: ");

    scanf("%d", &binary);

    printf("Decimal Equivalent: %d", binaryToDecimal(binary));

    return 0;
}
```

Decimal to Binary Conversion

```

#include <stdio.h>

void decimalToBinary(int n) {

    int binary[32], i = 0;

    while (n > 0) {

        binary[i++] = n % 2;

        n /= 2;

    }

    printf("Binary Equivalent: ");

    for (int j = i - 1; j >= 0; j--) {

        printf("%d", binary[j]);

    }

}

int main() {

    int decimal;

    printf("Enter a decimal number: ");

    scanf("%d", &decimal);

    decimalToBinary(decimal);

    return 0;

}

```

Frequency of Characters

```

#include <stdio.h>

#include <string.h>

int main() {

    char str[100], freq[256] = {0};

```

```
printf("Enter a string: ");

gets(str);

for (int i = 0; str[i] != '\0'; i++) {

    freq[(int)str[i]]++;

}

printf("Character Frequency:\n");

for (int i = 0; i < 256; i++) {

    if (freq[i] != 0) {

        printf("%c: %d\n", i, freq[i]);

    }

}

return 0;

}
```